

Vision-Based 2D Scan Generation for Obstacle Avoidance Using Floor Segmentation

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Abstract: This extended abstract presents a monocular virtual scan generation method for indoor autonomous mobile robots. The proposed system segments floor regions from RGB camera images and converts non-floor pixels into approximate 2D scan observations through camera-calibrated lookup tables. The generated scan is published in a ROS-compatible range format and integrated with the Nav2 navigation stack for local obstacle avoidance. Experiments in a structured indoor corridor show that the method provides useful, but limited, obstacle cues without a dedicated LiDAR sensor. In controlled range-estimation trials, the absolute error remained within 0.261 m under static conditions. In a 10 m static frontal obstacle avoidance experiment, the robot completed 39 of 50 trials. These results suggest that monocular scan generation can be a lightweight sensing option for constrained indoor environments, while robustness under reflective floors, illumination changes, and broader obstacle conditions remains future work.

Keywords: Floor segmentation, monocular camera, virtual scan, obstacle avoidance, ROS 2, Nav2, indoor mobile robot.

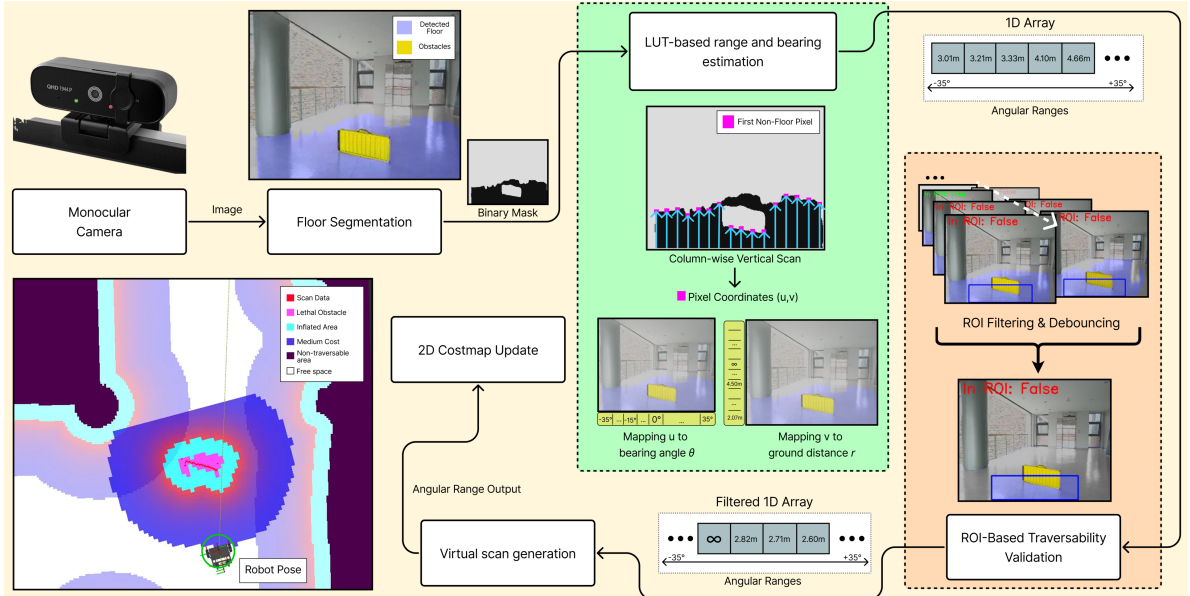


Fig. 1. Overview of the proposed virtual scan pipeline.

1. INTRODUCTION

Autonomous mobile robots (AMRs) are increasingly used in structured indoor environments, but robust perception remains a practical deployment challenge. LiDAR and other active range sensors provide direct geometric measurements, yet they increase hardware cost, payload, power consumption, and integration complexity [1], [2]. Cameras are lightweight and widely available, but visual outputs are not naturally expressed in the planar range format expected by many navigation pipelines.

This work addresses that interface gap by converting monocular floor-segmentation results into virtual 2D scan observations. Rather than claiming LiDAR-level ranging performance, the goal is to determine whether

a camera-only pipeline can provide useful obstacle cues for constrained indoor navigation. The main contribution is a practical pipeline that maps floor masks into laser-scan-like measurements and feeds them to ROS 2/Nav2 costmaps for real-time obstacle avoidance.

2. METHOD

Fig. 1 summarizes the proposed system. A forward-facing monocular camera captures RGB frames, and a lightweight YOLOv11n-seg model is fine-tuned to segment the floor region [3]. Pixels classified as floor are treated as traversable, whereas non-floor pixels inside a predefined frontal region of interest are treated as potential obstacle evidence.

To generate a scan-like representation in real time,

the system precomputes two lookup tables from the calibrated camera geometry, following the range-observation abstraction commonly used in probabilistic robotics and occupancy-grid mapping [4], [5]. A column-to-bearing table maps each image column to an angular scan bin, and a row-to-range table maps image rows to approximate ground-plane distance under a fixed camera height, fixed pitch angle, and locally flat-floor assumption. During operation, the lowest non-floor pixel in each column is selected as the approximate obstacle-floor contact point. Its image coordinates are converted into an angular bin and range estimate using the lookup tables, and the nearest range is retained for each scan bin.

The resulting one-dimensional scan is published in a range-message-compatible form and consumed by Nav2 as local obstacle information [6], [7]. A simple ROI-based floor-visibility check is also used to reduce stale obstacle artifacts: when the frontal region remains consistently clear, the system can trigger costmap clearing. The method therefore assumes a structured indoor environment with a flat floor and a fixed camera mounting configuration; changes in camera pose or floor geometry require recalibration.

3. EXPERIMENTAL RESULTS

Two experiments were conducted. First, range-estimation performance was evaluated under controlled static conditions at a reference distance of 2.50 m and three bearing conditions. The observed absolute error remained within 0.261 m, indicating that the scan is not suitable for precise ranging but can provide coarse proximity cues for local navigation.

Second, the method was integrated with a mobile robot running monocular SLAM and Nav2. The robot was commanded to traverse a 10 m straight path with a single rectangular cardboard-box obstacle placed at the center of the path. Across 50 trials, the robot completed 39 successful runs, corresponding to a 78% success rate. During successful trials, obstacle cues were generated online, represented in the local costmap, and used by Nav2 to produce an avoidance path.

The main failure mode was segmentation-induced obstacle jitter, especially under a glossy floor and dynamic illumination. Eight of the 11 failed trials were attributed to unstable obstacle position estimates from floor-segmentation noise, and three occurred when the robot returned to the nominal path too quickly after avoidance. These results support further investigation of monocular virtual scanning as a lightweight sensing option, but they also show that stronger target-domain segmentation data and temporal filtering are needed.

4. CONCLUSION

This extended abstract presented a vision-based 2D scan generation method that converts monocular floor

segmentation into scan-like obstacle observations for ROS 2/Nav2 navigation. The approach demonstrated useful obstacle-awareness behavior in a constrained indoor scenario, including repeated static obstacle avoidance without a dedicated LiDAR sensor under controlled indoor conditions. Future work will improve floor-mask stability, expand target-domain training data, evaluate additional obstacle and lighting conditions, and release the ROS 2 Humble virtual scan generator package after publication.

ACKNOWLEDGEMENT

This research was supported by the ANCHOR program Glocal University 30 through the Gyeongbuk ANCHOR CENTER, funded by the Ministry of Education (MOE) and the Gyeongsangbuk-do, Republic of Korea. (2026-ANCHOR-15-119)

USE OF AI TOOLS

ChatGPT by OpenAI and Codex by OpenAI were used to assist with $\LaTeX$ editing and composition, translation, and grammar correction. All scientific content, experimental results, interpretations, references, and the final manuscript were reviewed and verified by the authors, who take full responsibility for the content of this work.

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